

Mathematical background

- In order to understand some of cryptographic algorithms dealt with throughout this course, it is necessary to have some background in two areas of mathematics
 1. Number Theory.
 2. Abstract Algebra.
- New Advanced Encryption Standard (AES) relies on the subject of finite fields which forms a part of abstract algebra.
- Going to deal with Number Theory for the moment.

Number Theory

- Number theory deals with the theory of numbers and is probably one of the oldest branches of mathematics.

- It is divided into several areas including elementary, analytic and algebraic number theory.
- These are distinguished more by the methods used in each than the type of problems posed.
- Relevant ideas discussed here and include:
 - prime numbers
 - the greatest common divisor
 - the modulus operator
 - the modular inverse

Prime numbers

- A prime number p is an integer greater than 1 with only two positive divisors, 1 and itself.
- Therefore its entire set of divisors (i.e. its factors) consist only of four integers ± 1 and $\pm p$

- It can be seen that 1 is not a prime number.
- Prime numbers are of the utmost importance to certain cryptographic algorithms and most of the techniques used will not work without them.
- Any positive integer $I \geq 2$ is either a prime or can be expressed as the product of primes.
- This is known as the fundamental theorem of arithmetic:

$$I = P_N^{\epsilon_N} \times P_{N-1}^{\epsilon_{N-1}} \times \dots \times P_1^{\epsilon_1}, \quad P_N > P_{N-1} > \dots > P_1$$

- - - (i)

- Another way of looking at this would be:

$$I = \prod_s P_n^{\epsilon_n} \quad \epsilon_n \geq 0 \quad - \dots \text{ (ii)}$$

Here 's' is the set of all prime numbers.

- In general most of the exponents ϵ_n will be '0'.

- As a result of equation (i) and (ii) any integer > 1 , that is not a prime number is known as a composite number.
- It can be seen that from this and the definition of a prime number above, that 1 is neither prime nor composite.
- The first prime numbers are:
2, 3, 5, 7, 11, 13, 17, 19, 23 and 29.

Division

- Any integer can be expressed as $n = q \times m + r$, where n, q , and r are integers, m is a positive integer and $0 \leq r < m$
- The remainder (also known as residue) r , must be non-negative (i.e. either positive or 0)

- This is seen by two restrictions:
 1. $0 \leq r < m$
 2. $q = \left\lfloor \frac{n}{m} \right\rfloor$
- The notation $\lfloor x \rfloor$ is known as the floor of the integer x and is the greatest integer $\leq x$.
- The notation $\lceil x \rceil$ is the ceiling of the integer x and is the least integer $\geq x$.
- For example, $24 \div 10$ is 2 with a remainder 4 however, $-24 \div 10$ is -3 with a remainder 6 and not -2 with a remainder -4 as might be expected.
- If $r=0$ then n is said to be a multiple of m . This is also the same as saying that m divides n , is a divisor of n or is a factor of n and the notation used to express this is $m|n$.

- The greatest common divisor, $m_{\max}$, of two integers a and b is the largest positive integer that will divide both a and b without a remainder.
- Therefore $m_{\max} | a$, $m_{\max} | b$ and $m_n | m_{\max}$ for any divisor m_n of a and b .
- The notation generally used to represent this is $\text{gcd}(a, b) = m_{\max}$.
- If $\text{gcd}(a, b) = 1$, this means that a and b have no common factors other than 1.
- Such pair is known as relatively prime or co-prime
- Along with prime numbers, numbers that are relatively prime have considerable importance in cryptography as will be seen later.

- The greatest common divisor of two positive integers a and b ($\text{gcd}(a, b)$) can be determined by a procedure known as Euclid's Algorithm. It is based on the theorem that
$$\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$$

- The expression 'mod' is used in modular arithmetic which is a special kind of arithmetic involving remainders as will be seen next.

Modular Arithmetic

- The symbol used ($\equiv$) is known as the congruence symbol.
- Modular relationships are of the form $n \equiv R \pmod{m}$ (spoken as "n is congruent to R mod m") where n and R are integers and m is a positive integer known as the modulus.